

Chapter-2

The Cells of the Nervous Systems & Communication Between Within Neurons

1st year BSc Psychology, BATCH – 2026-29





Anatomy of Neurons & Glia

Adult human brain contains approximately 100 billion neurons. The exact number varies from person to person.

- **Nervous system** □ A network of cells and organs that receives information, processes it, and sends messages throughout the body.
- **Nerve** □ a bundle of many axons from different neurons that carry signals between the brain/spinal cord and the rest of the body.

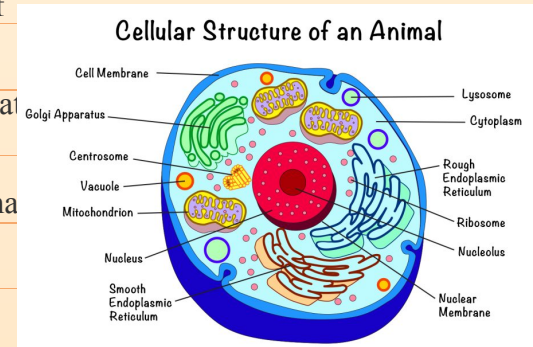
The nervous system consists of two kinds of cells:

- A **neuron** is a specialized nerve cell that receives, processes, and transmits information throughout the body using electrical and chemical signals.
- **Glia** □ Also known as “neuroglia”. These are non-neuronal cells that support and protect the neurons of the brain, produce myelin, provide nutrients, supply oxygen, remove waste, repair damages, etc.
- **Before late 1800s** □ microscopic views revealed little detail about the brain. Observers noted long, thin fibers between one neuron’s cell body and another, but they could not see whether each fiber merged into the next cell or stopped before it
- Then, **in the late 1800s** □ **Santiago Ramón y Cajal** (Spanish neuroscientist who is often called the “**Father of Modern Neuroscience**”) used newly developed **staining techniques** to show that a small gap separates the tips of one neuron’s fibers from the surface of the next neuron.
- Because of Cajal's work, scientists concluded that the **brain is made of billions of separate neurons** that communicate across tiny gaps called **synapses**.

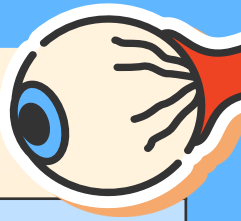


Structures of an Animal Cell

- Neurons have much in common with the rest of the body's cells.
- **Cell membrane (plasma membrane)** □ Surface of a cell. This structure separates the inside of the cell from the outside environment. It is composed of two layers of fat molecules that are free to flow around one another. Most chemicals cannot cross the membrane, but specific protein channels in the membrane permit a controlled flow of water, oxygen, sodium, potassium, calcium, chloride, and other important chemicals.
- Except for mammalian red blood cells, all animal cells have a **nucleus** □ structure that contains the chromosomes & DNA.
- **Mitochondria** □ structure that performs metabolic activities, providing the energy that the cell requires for all other activities. Mitochondria require glucose and oxygen to function.
- **Ribosomes** □ sites at which the cell synthesizes new protein molecules.
- **Proteins** □ provide building materials for the cell and facilitate various chemical reactions.
- **Endoplasmic reticulum** □ a network of thin tubes that transport newly synthesized proteins to other locations.



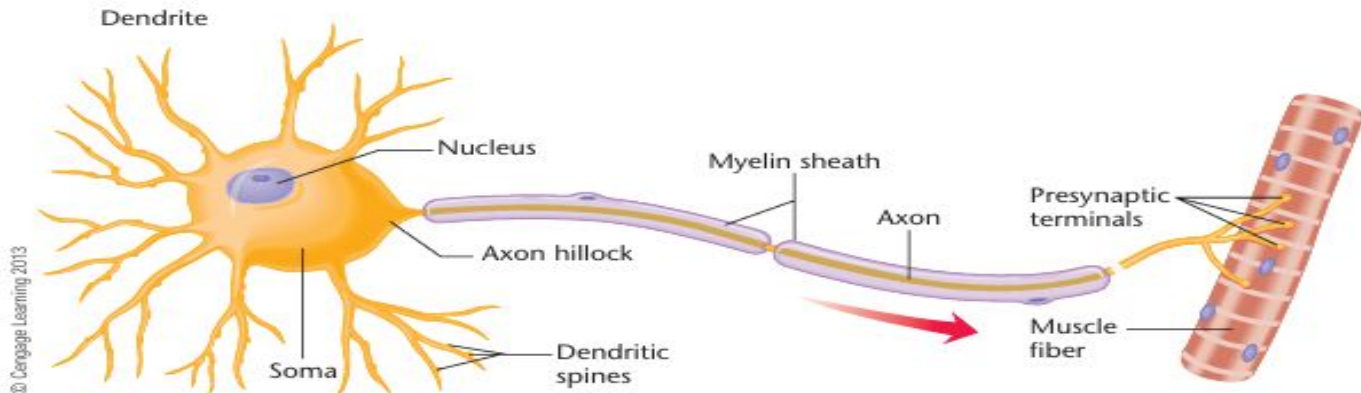
Structure of a Neuron



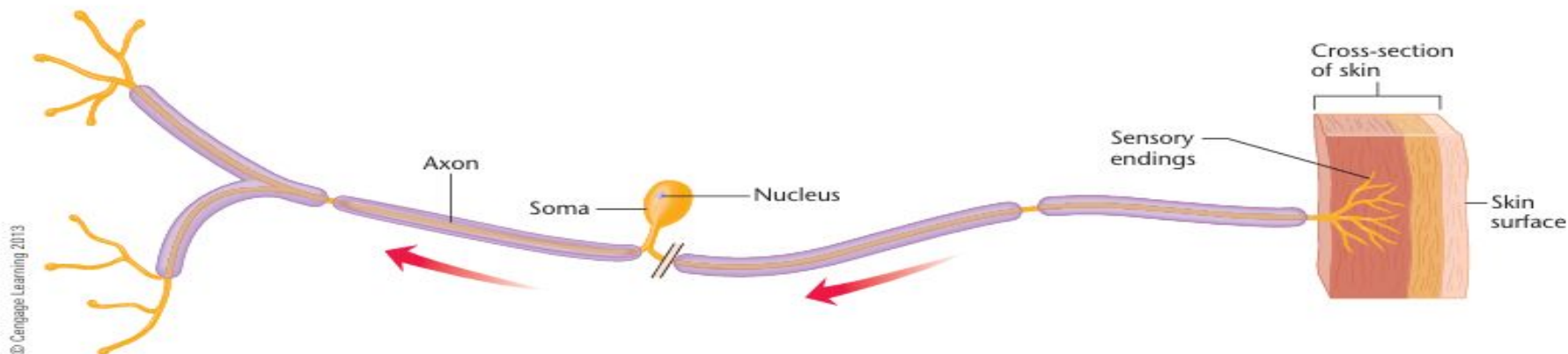
PARTS OF A NEURON	DESCRIPTION	FUNCTION
Dendrites	Branching fibers that get narrower near their ends. (Dendrite is a greek root word meaning "tree." A dendrite branches like a tree.)	Receives information from other neurons
Dendritic spines	Short out growths that increase the surface area available for synapses.	Increase the number of synaptic connections, allowing the neuron to receive more information.
Soma (Cell Body)	Contains the nucleus, ribosomes, and mitochondria; the metabolic center of the neuron.	Maintains the neuron, integrates incoming signals, and supports cell metabolism.
Nucleus	Membrane-bound organelle containing the cell's DNA (genetic material).	Controls the neuron's activities by regulating gene expression and protein synthesis.
Axon	A long, thin fiber of constant diameter that extends from the soma.	Carries nerve impulses (action potentials) away from the cell body to other neurons, muscles, or glands.
Axon Hillock	Cone-shaped region where the soma joins the axon.	Integrates incoming signals and initiates the action potential if the threshold is reached.
Myelin Sheath	Fatty insulating covering around many axons, formed by schwann cells (PNS) or oligodendrocytes (CNS).	Insulates the axon and greatly increases the speed of nerve impulse conduction.
Nodes of Ranvier	Small gaps between adjacent myelin sheath segments along the axon.	Allow the action potential to "jump" from node to node (saltatory conduction), increasing the speed of nerve impulse transmission.
Presynaptic terminal/ end bulb/ bouton	Bulb-like endings at the branches of an axon containing synaptic vesicles filled with neurotransmitters.	Releases neurotransmitters to transmit information to the next neuron, muscle, or gland across the synapse.

FIGURE 2.5 The components of a vertebrate motor neuron

The cell body of a motor neuron is located in the spinal cord. The various parts are not drawn to scale; in particular, a real axon is much longer in proportion to the soma.



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FIGURE 2.6 A vertebrate sensory neuron

Note that the soma is located on a stalk off the main trunk of the axon. (As in Figure 2.5, the various structures are not drawn to scale.)

Types of Neurons Based on Function

01

Sensory neurons

Specialized at one end to be highly sensitive to a specific stimulation, such as light, sound, or touch. This is a neuron conducting touch information from the skin to the spinal cord.



02

Motor neurons

Carries nerve impulses from the CNS (brain or spinal cord) to muscles or glands, causing them to respond.

03

Interneurons/ Intrinsic neurons

Processes information and connects neurons within the CNS

Variations among neurons

- Neurons vary enormously in size, shape, and function.
- The function of a neuron relates to its shape and number of dendrites.

The diverse shapes of neurons are as follows:

- **Purkinje cell (Cerebellum):** Has many branching dendrites, so it receives signals from many neurons.
- **Bipolar cell (Retina):** Has short dendrites, so it receives signals from only a few neurons.
- **Sensory neuron (Skin □ Spinal cord):** Carries touch, pain, and temperature information from the skin to the spinal cord.
- **Pyramidal cell (Motor cortex):** Sends signals from the brain to help control voluntary movement.
- **Kenyon cell (Honeybee brain):** Processes sensory information and helps with learning and memory.

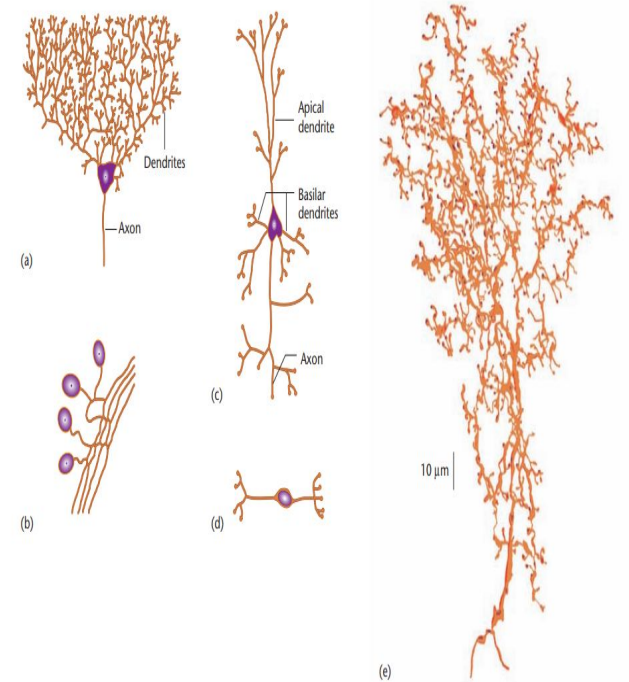


FIGURE 2.9 The diverse shapes of neurons

(a) Purkinje cell, a cell type found only in the cerebellum; (b) sensory neurons from skin to spinal cord; (c) pyramidal cell of the motor area of the cerebral cortex; (d) bipolar cell of retina of the eye; (e) Kenyon cell, from a honeybee. (Source: Part e courtesy of R. G. Goss)

Variations among Glial Cells

- **Glia (neuroglia)** □ **Support and protect neurons** but do not transmit information over long distances like neurons do.
- The word *glia* means “**glue**” in Greek. Early scientists thought glia were like glue that held neurons together.
- Glial cells are **smaller and more numerous** than neurons.

The brain and nervous system have **different types of glia**, and each has a different function.

1. **Astrocytes**

- **Star-shaped cells.**
- They surround the **presynaptic terminals** of groups of neurons.
- They help **control ions** around neurons and can help synchronize the activity of neurons.
- They **remove waste** produced by damaged or dead neurons.
- They help **control blood flow** in different brain areas.
- When a brain area becomes very active, astrocytes help **dilate blood vessels**, bringing more **oxygen and nutrients** to that area.
- Astrocytes also help **recycle glutamate**. They take up excess glutamate, convert it into **glutamine**, and return it to neurons. Neurons can then convert it back into glutamate.

Variations among Glial Cells

• 2. Microglia

- They are **very small cells**.
- They remove **dead cells and waste**.
- They also help fight **viruses, fungi, and other microorganisms**.
- Therefore, microglia act like the **immune system of the brain**.

3. Oligodendrocytes

- Found in the **CNS (brain and spinal cord)**.
- They produce **myelin sheaths** around axons.
- Myelin **insulates axons and helps signals travel faster**.

4. Schwann cells

- Found in the **PNS (outside the brain and spinal cord)**.
- They also produce **myelin sheaths** around axons

5. Radial glia

- They guide the **migration of neurons** and their axons and dendrites **during embryonic development**.
- **When embryological development finishes, most radial glia differentiate into neurons, and a smaller number differentiate into astrocytes and oligodendrocytes.**

NEURONS VERSUS GLIAL CELLS	
NEURONS	GLIAL CELLS
The specialized cells of the nervous system, receiving and transmitting chemical or electric signals	The cells that surround neurons, providing support and insulating them
The structural and functional units of the nervous system	The non-neuronal cells of the nervous system
Responsible for the transmission of nerve impulses between the central nervous system and body	Maintain homeostasis, providing support and protection to the neurons
Directly take part in the signal transduction	Increase the transduction speed by myelination of axons
Types: Sensory, motor, and interneurons	Types: Oligodendrocytes, astrocytes, ependymal cells, microglia, Schwann cells, and satellite cells
Visit www.PEDIAA.com	

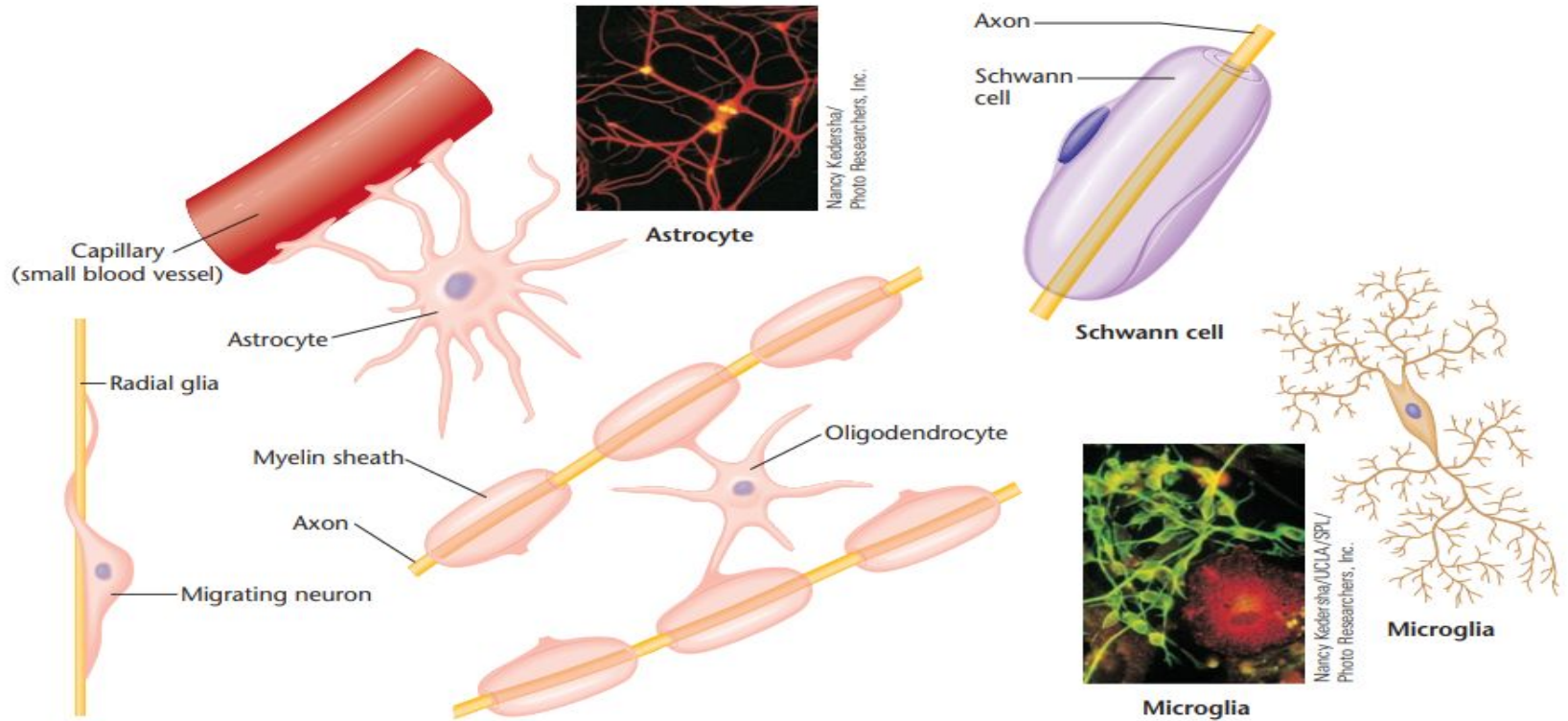


FIGURE 2.10 Shapes of some glia cells

Oligodendrocytes produce myelin sheaths that insulate certain vertebrate axons in the central nervous system; Schwann cells have a similar function in the periphery. The oligodendrocyte is shown here forming a segment of myelin sheath for two axons; in fact, each oligodendrocyte forms such segments for 30 to 50 axons. Astrocytes pass chemicals back and forth between neurons and blood and among neighboring neurons. Microglia proliferate in areas of brain damage and remove toxic materials. Radial glia (not shown here) guide the migration of neurons during embryological development. Glia have other functions as well. (© Cengage Learning 2013)

The Blood–Brain Barrier

- Although the brain, like any other organ, needs to receive nutrients from the blood, many chemicals cannot cross from the blood to the brain
- The mechanism that excludes most chemicals from the vertebrate brain is known as the **blood–brain barrier**.

a) Why We Need a Blood–Brain Barrier?

- When a virus invades a cell, mechanisms within the cell extrude virus particles through the membrane so that the immune system can find them. When the immune system cells identify a virus, they kill it and the cell that contains it.
- However, with few exceptions, the vertebrate brain does not replace damaged neurons.
- To minimize the risk of irreparable brain damage, the body builds a wall along the sides of the brain's blood vessels.
- This wall keeps out most viruses, bacteria, and harmful chemicals.

What happens if a virus does enter the nervous system?

- Certain viruses, such as the rabies virus, evade the blood–brain barrier, infect the brain, and lead to death. For several other viruses that enter the nervous system, microglia and other mechanisms attack the viruses or slow their reproduction without killing the neurons they invaded
- ✦ However, a virus that enters your nervous system probably remains with you for life. For example, the virus responsible for chicken pox and shingles enters spinal cord cells. No matter how effectively the immune system attacks that virus outside the nervous system, virus particles remain in the spinal cord, from which they can emerge decades later. The same is true for the virus that causes genital herpes

The Blood–Brain Barrier (Cont.)

b) How the Blood–Brain Barrier Works?

- The blood–brain barrier depends on the endothelial cells that form the walls of the capillaries
- Outside the brain, such cells are separated by small gaps, but in the brain, they are joined so tightly that virtually nothing passes between them.
- Barrier keeps out useful chemicals as well as harmful ones. Those useful chemicals include all fuels and amino acids, the building blocks for proteins. For the brain to function, it needs special mechanisms to get these chemicals across the blood–brain barrier

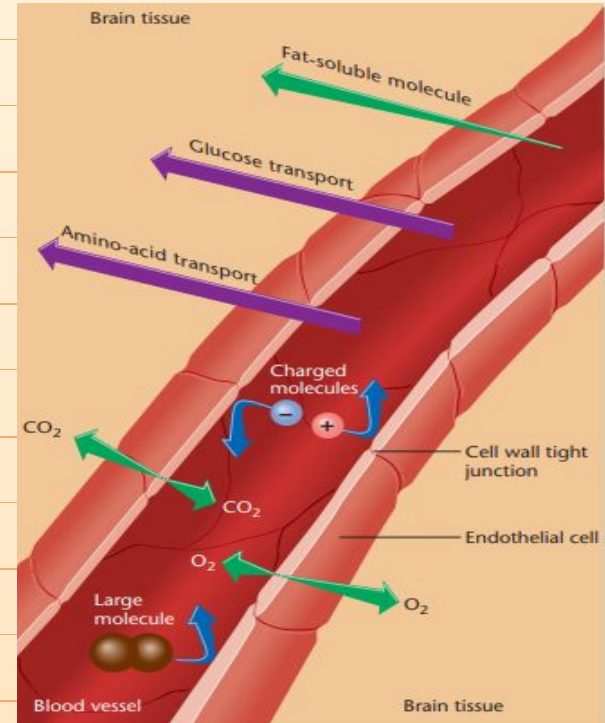
The brain has several such mechanisms.

- First, small uncharged molecules, including oxygen and carbon dioxide, cross freely. Water crosses through special protein channels in the wall of the endothelial cells.
- Second, molecules that dissolve in the fats of the membrane also cross passively. Examples include vitamins A and D and all the drugs that affect the brain—from antidepressants and other psychiatric drugs to illegal drugs such as heroin.
- For a few other chemicals, the brain uses active transport, a protein-mediated process that expends energy to pump chemicals from the blood into the brain.
- Chemicals that are actively transported into the brain include glucose (the brain's main fuel), amino acids (the building blocks of proteins), purines, choline, a few vitamins, iron, and certain hormones

The Blood–Brain Barrier (Cont.)

b) How the Blood–Brain Barrier Works? (Cont.)

- The blood–brain barrier is essential to health.
- In people with Alzheimer's disease or similar conditions, the endothelial cells lining the brain's blood vessels shrink, and harmful chemicals enter the brain
- However, the barrier also poses a difficulty in medicine because it keeps out many medications.
- Brain cancers are difficult to treat because nearly all the drugs used for chemotherapy fail to cross the blood–brain barrier

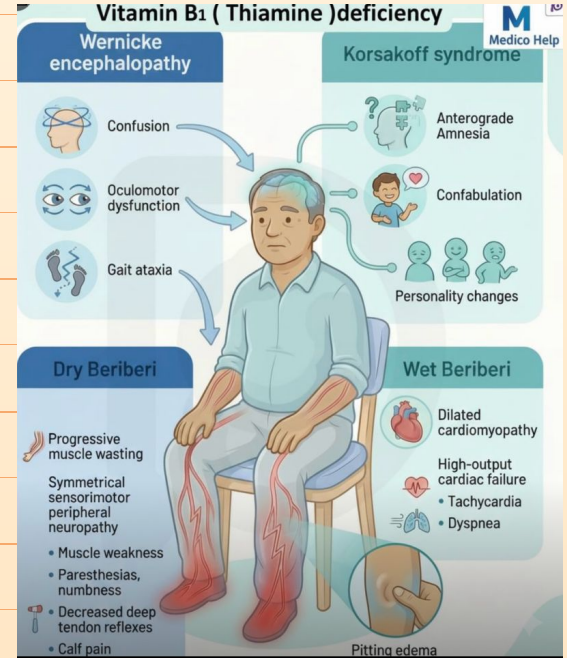


Nourishment in Vertebrate Neurons

- Most cells use a variety of carbohydrates and fats for nutrition, but vertebrate neurons depend almost entirely on glucose, a sugar. **But why?**
- Because the metabolic pathway that uses glucose requires oxygen, neurons need a steady supply of oxygen. The brain uses about 20% of all the oxygen consumed in the body.

Why do neurons depend so heavily on glucose?

- Although neurons have the enzymes necessary to metabolize other fuels, glucose is practically the only nutrient that crosses the blood-brain barrier after infancy, except for ketones (a kind of fat), and ketones are seldom available in large amounts
- Although neurons require glucose, glucose shortage is rarely a problem.
- The liver makes glucose from many kinds of carbohydrates and amino acids, as well as from glycerol, a breakdown product from fats.
- The **only likely problem is an inability to use glucose.**
- To use glucose, the body needs **vitamin B1, thiamine.**
- Prolonged thiamine deficiency**, common in chronic alcoholism, leads to death of neurons and a condition called **Korsakoff's syndrome**, marked by severe memory impairments

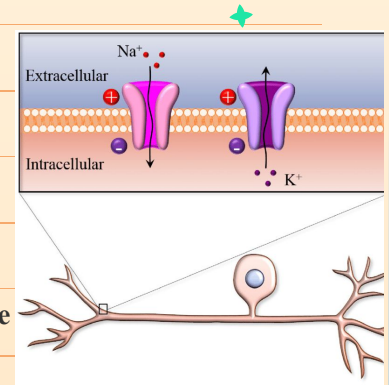




The Nerve Impulse

The Resting Potential of the Neuron

- All parts of a neuron are covered by a **cell membrane** about **8 nanometers (nm)** thick, **composed of two layers** called **phospholipid bilayer** (an inner layer and an outer layer) of phospholipid molecules (containing chains of fatty acids and a phosphate group).
- Embedded among the phospholipids are **cylindrical protein molecules** through which various chemicals can pass
- Thus, **cell membrane controls the flow of chemicals between the inside and outside of the cell.**
- In the absence of any outside disturbance, the membrane maintains an electrical gradient, also known as **polarization**—**a difference in electrical charge between the inside and outside of the cell.**
- Inside the neuron, there are **negatively charged proteins**. These contribute to the **inside being more negative than the outside.**
- This **difference in actual voltage in a resting neuron** is called the **resting potential.**
- *Researchers measure the resting potential by inserting a very thin microelectrode into the cell body. The diameter of the electrode must be as small as possible so that it enters the cell without causing damage. The most common electrode is a fine glass tube filled with a concentrated salt solution and tapering to a tip diameter of 0.0005 mm or less. A reference electrode outside the cell completes the circuit. Connecting the electrodes to a voltmeter, we find that the neuron's interior has a negative potential relative to its exterior. A typical level is -70 millivolts (mV), but it varies from one neuron to another.*





The Nerve Impulse



a) Forces Acting on Sodium and Potassium Ions

- ❑ If charged ions could flow freely across the membrane, the membrane would **depolarize** (inside becomes less negative).
- ❑ However, the **membrane is selectively permeable** (some chemicals like oxygen, carbon dioxide, urea, and water pass through it more freely than large or electrically charged ions and molecules).
- ❑ A few biologically important ions, such as **sodium, potassium, calcium, and chloride**, **cross through membrane channels (or gates) that are sometimes open and sometimes closed**.
- ❑ When the membrane is at rest ☐ the sodium channels are closed, preventing almost all sodium flow & potassium channels are nearly but not entirely closed, so potassium flows slowly.
- ❑ Stimulation opens the channels more widely.
- ❑ **The sodium-potassium pump**, a protein complex, **repeatedly transports 3 sodium ions out of the cell while drawing 2 potassium ions into it**.
- ❑ The sodium-potassium pump is **an active transport that requires energy (ATP)**.
- ❑ As a result of the sodium-potassium pump, **sodium ions are more than 10 times more concentrated outside the membrane than inside, and potassium ions are similarly more concentrated inside than outside**.
- ❑ The sodium-potassium pump is **effective only because of the selective permeability of the membrane, which prevents the sodium ions that were pumped out of the neuron from leaking right back in** again.
- ❑ When sodium ions are pumped out, they stay out. However, some of the potassium ions pumped into the neuron slowly leak out, carrying a positive charge with them. That leakage increases the electrical gradient across the membrane.

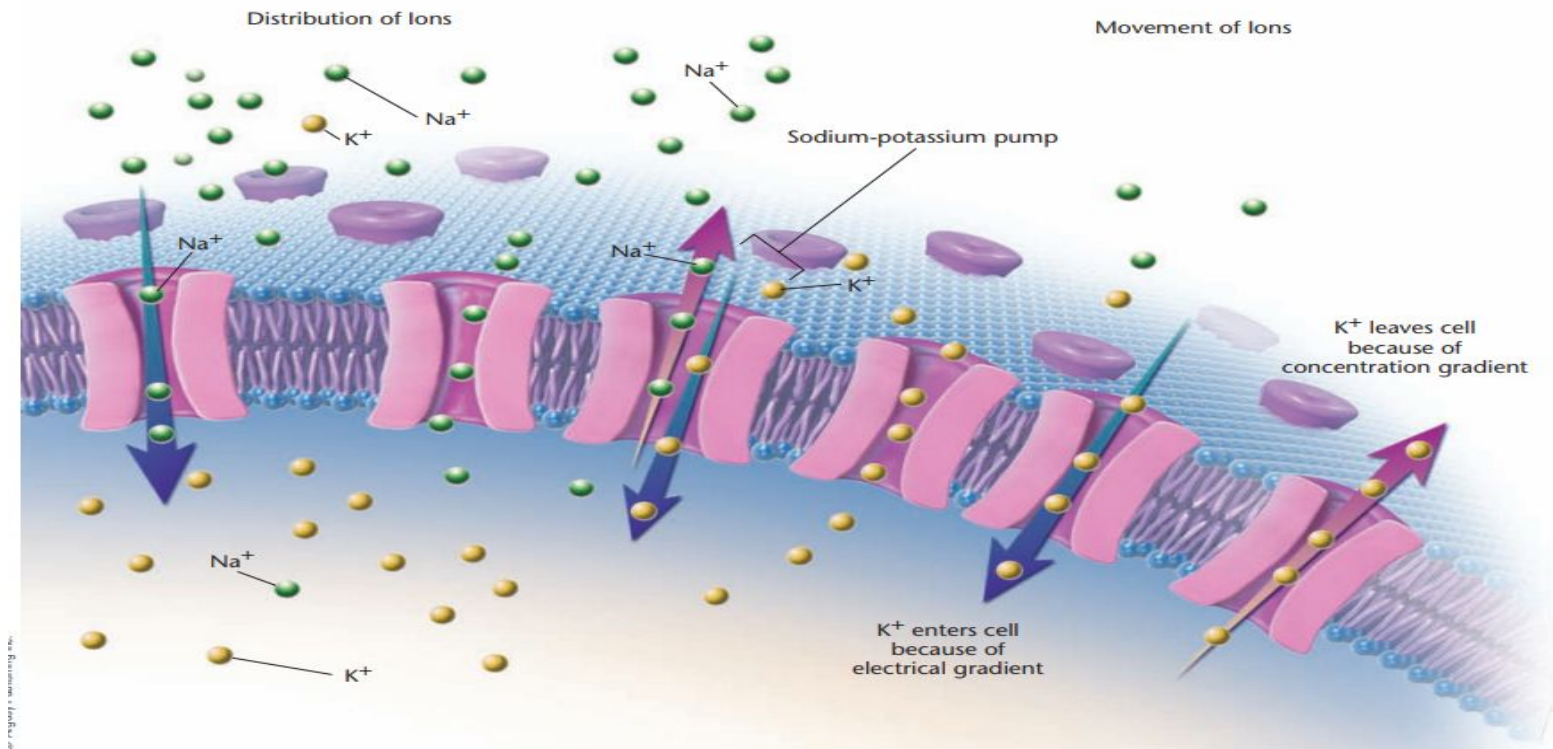


FIGURE 2.15 The sodium and potassium gradients for a resting membrane

Sodium ions are more concentrated outside the neuron. Potassium ions are more concentrated inside. Protein and chloride ions (not shown) bear negative charges inside the cell. At rest, very few sodium ions cross the membrane except by the sodium-potassium pump. Potassium tends to flow into the cell because of an electrical gradient but tends to flow out because of the concentration gradient. However, potassium gates retard the flow of potassium when the membrane is at rest.



The Nerve Impulse

a) Forces Acting on Sodium and Potassium Ions (Cont.)

- When the neuron is at rest, two forces act on **sodium**, both tending to push it into the cell.

Electrical gradient (the difference in electrical charge across membrane)

- Sodium is positively charged and the inside of the cell is negatively charged. Opposite electrical charges attract, so the **electrical gradient tends to pull sodium into the cell.**

Concentration gradient (the difference in distribution of ions across the membrane)

- Sodium is more concentrated outside than inside, so just by the laws of probability, **sodium is more likely to enter the cell**
- Given that both the electrical gradient and the concentration gradient tend to move sodium ions into the cell, sodium would move rapidly if it could. However, the sodium channels are closed when the membrane is at rest, and almost no sodium flows except for the sodium pushed out of the cell by the sodium-potassium pump.
- **Potassium is subject to competing forces.**
- Potassium is positively charged and the inside of the cell is negatively charged, so the **electrical gradient tends to pull potassium in.**
- However, potassium is more concentrated inside the cell than outside, so the **concentration gradient tends to drive it out.**
- If the potassium channels were wide open, potassium would have a small net flow out of the cell. That is, the electrical gradient and concentration gradient for potassium are almost in balance, but not quite. The sodium-potassium pump pulls more potassium into the cell as fast as it flows out of the cell, so the two gradients cannot get completely in balance.
- ✕ The cell has negative ions, too. Negatively charged proteins inside the cell are responsible for the membrane's polarization. Chloride ions, being negatively charged, are mainly outside the cell. **When the membrane is at rest, the concentration gradient and electrical gradient balance, so opening chloride channels produces little effect.** However, chloride does have a net flow when the membrane's polarization changes.



The Nerve Impulse

b) Why a Resting Potential?

- ❑ The body invests much energy (ATP) to operate the sodium–potassium pump, which maintains the resting potential.

Why spend so much energy? Why not wait until the neuron is stimulated and then prepare it to respond?

- ❑ The resting potential prepares the neuron to respond rapidly.
- ❑ Excitation of the neuron opens channels that allow sodium to enter the cell rapidly. Because the membrane did its work in advance by maintaining the concentration gradient for sodium, the cell is prepared to respond vigorously to a stimulus.
- ❑ **Example:** Compare the resting potential of a neuron to a poised bow and arrow: An archer who pulls the bow in advance and then waits is ready to fire at the appropriate moment. The neuron uses the same strategy.
- ❑ The resting potential remains stable until the neuron is stimulated.
- ❑ <https://youtube.com/shorts/4ddZBaUqEko?si=Q-Mp5Ax4JA0GOgqo>

The Action Potential

- **Polarization** □ condition of **having different charges across the membrane** (*more negative inside and more positive outside*)
- **Resting potential** □ **measured voltage difference when the neuron is at rest.** (*inside = -70 mv; outside = 0 mv*)
- **Action potential** □ **electrical signal/message generated and propagated along the axon because of a rapid change in the membrane potential when the neuron fires**
(*depolarization □ repolarization □ hyperpolarization □ refractory period □ return toward resting potential*)

Action Potential Steps

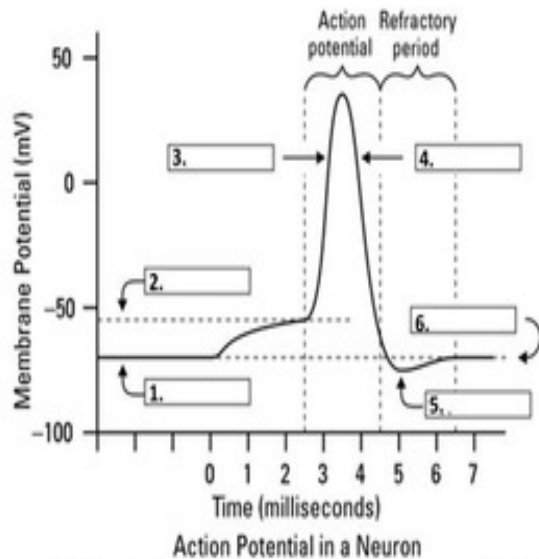
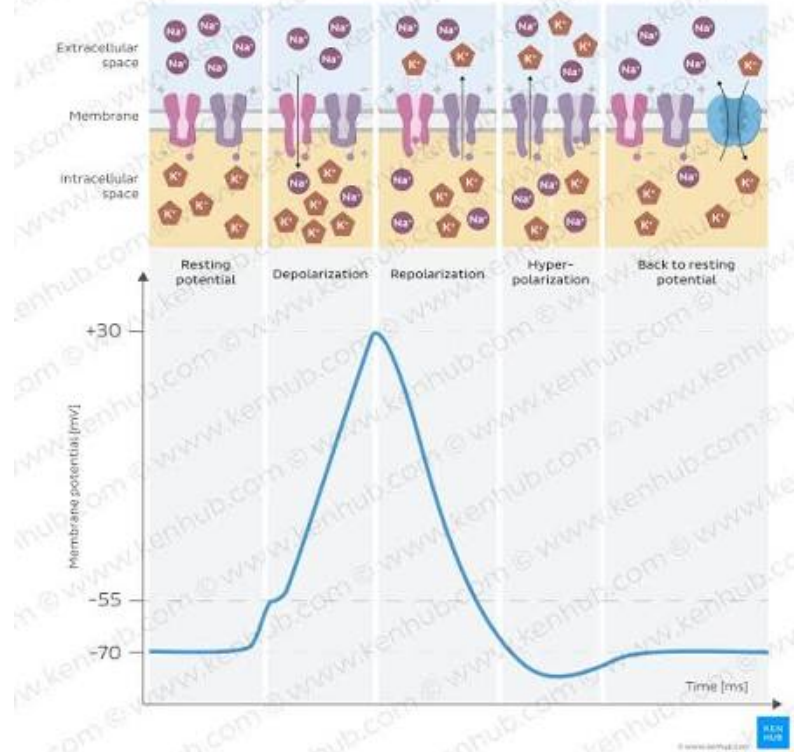


Image: <https://www.dummies.com/education/science/biology/action-potential-of->

1. Resting Membrane Potential (RMP) = -70 mV
2. Threshold = -55 mV
3. **Depolarization** = Voltage-gated Na⁺ Open (Na⁺ FLOWS IN)
4. **Repolarization** = Voltage-gated K⁺ Open (K⁺ FLOWS OUT)
5. **Hyperpolarization** = Overshoot, K⁺ channels close
6. **Na⁺/K⁺ pump** quickly re-establishes RMP (-70mV)

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The Action Potential

The Molecular Basis of the Action Potential

- A neuron's membrane contains several types of cylindrical proteins that can open or close.
- **When one of these proteins is open, it allows a particular type of ion to cross the membrane.**
- A protein that allows sodium to cross is called a **sodium channel**, one that allows potassium to cross is a **potassium channel**, and so forth.
- The ones regulating sodium and potassium are **voltage-gated channels**. That is, their permeability depends on the voltage difference across the membrane (When the membrane voltage changes, these channels open or close, changing how easily Na^+ or K^+ can cross the membrane.)
- **At the resting potential**, the sodium channels are closed (permitting no sodium to cross) and the **potassium channels are almost closed** (allowing only a little flow of potassium).
- As the membrane becomes **depolarized** (inside of the neuron becomes less negative compared with its resting state), both the **sodium and the potassium channels begin to open**, allowing freer flow. At first, opening the potassium channels makes little difference, because the concentration gradient and electrical gradient are almost in balance anyway. However, opening the sodium channels makes a big difference, because both the electrical gradient and the concentration gradient tend to drive sodium ions into the neuron. When the **depolarization reaches the threshold** (minimum level of membrane depolarization needed to trigger an action potential) of the membrane, the **sodium channels open wide** enough for sodium to flow freely.

The Action Potential

The Molecular Basis of the Action Potential (Cont.)

- Driven by both the concentration gradient and the electrical gradient, the sodium ions enter the cell rapidly, until the electrical potential across the membrane passes beyond zero to a reversed polarity.
- At the peak of the action potential, the sodium gates snap shut and resist reopening for the next millisecond.
- After so many sodium ions have crossed the membrane, the inside of the cell has a slight positive charge instead of its usual negative charge.
- At this point both the concentration gradient and the electrical gradient drive potassium ions (+ve charged) out of the cell causing the inside of the cell to become more negative as **anions (-ve charged ions) remain inside** □ **repolarization** (membrane becomes negative again after becoming positive)
- Because the potassium channels remain open after the sodium channels close, enough potassium ions leave to drive the membrane beyond its usual resting level to a temporary **hyperpolarization**.
- At the end of this process, the membrane has returned to its resting potential, but the inside of the neuron has slightly more sodium ions and slightly fewer potassium ions than before.
- Thus, the **sodium-potassium pump** restores the original distribution of ions, but that process takes time.

The Action Potential

The Molecular Basis of the Action Potential (Cont.)

- After an unusually rapid series of action potentials, the pump cannot keep up with the action, and sodium accumulates within the axon.
- **Excessive buildup of sodium can be toxic to a cell.** (Excessive stimulation occurs only under abnormal conditions, however, such as during a stroke or after the use of certain drugs)
- **Local anaesthetic drugs**, such as Novocain and Xylocaine, attach to the sodium channels of the membrane, preventing sodium ions from entering, and thereby stopping action potentials
- When a dentist administers Novocain before drilling into one of your teeth, your receptors are screaming, “pain, pain, pain!” but the axons can’t transmit the message to your brain, and so you don’t feel it.

ACTION POTENTIAL

HOW NERVE SIGNALS TRAVEL

How electrical signals travel along neurons through changes in membrane potential. Designed for a global audience with advanced scientific clarity and excellent readability.

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What Is an Action Potential?

A rapid electrical signal that allows nerve cells using ourving nerve cells to communicate.

Resting Potential

The negative internal charge due to ion increased ion distribution; changes in membrane potential.

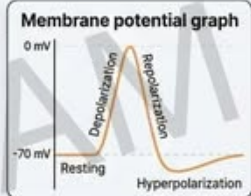
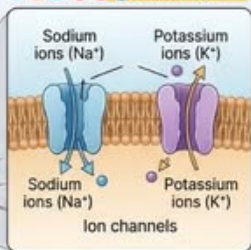
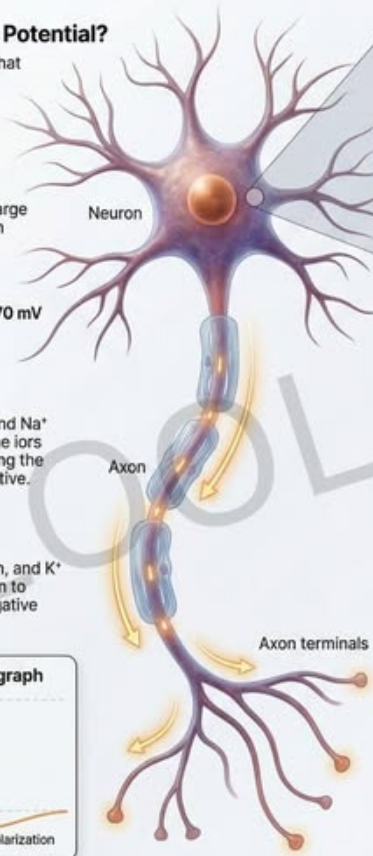
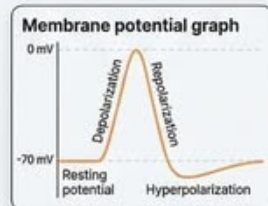
Resting membrane -70 mV

Depolarization

Sodium channels open and Na⁺ ions rush into the cell. The ions then unato the cell, making the membrane potential positive.

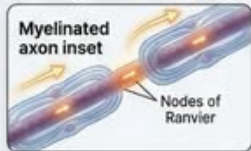
Repolarization

Potassium channels open, and K⁺ ions exit then the nes opn to the neuron, restorins negative membrane potential.



Saltatory Conduction

In myelinated axons, the electrical signal jumps between nodes of Ranvier, asaltatory conduction, increasing transmission speed.



Neural Communication

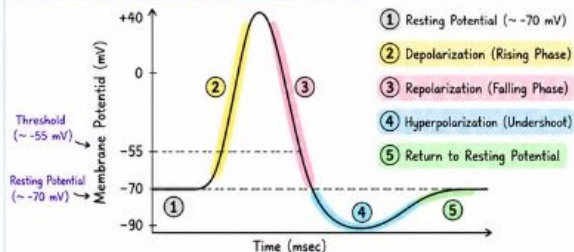
Action potentials allow neurons to transmit signals to muscles, glands, and other neurons.

★ NERVE ACTION POTENTIAL (N.A.P.) ★

1. KEY PLAYERS: IONS & GATED CHANNELS

- Na⁺ (Sodium): High concentration outside the cell; wants in. (Voltage-gated Na⁺ channels)
- K⁺ (Potassium): High concentration inside the cell; wants out. (Voltage-gated K⁺ channels)
- Na⁺/K⁺ Pump: Actively maintains gradients. (3 Na⁺ out, 2 K⁺ in using ATP)

2. THE ACTION POTENTIAL (GRAPH & PHASES)



- 1 Resting Potential (~-70 mV):**
 - Voltage-gated Na⁺ and K⁺ channels are closed.
 - K⁺ leak channels are open → K⁺ diffuses out.
- 2 Depolarization (Rising Phase):**
 - At threshold (~-55 mV), voltage-gated Na⁺ channels open rapidly.
 - Na⁺ rushes in → Membrane potential becomes positive (up to ~+30 to +40 mV).
- 3 Repolarization (Falling Phase):**
 - Na⁺ channels inactivate.
 - Voltage-gated K⁺ channels open → K⁺ flows out → Membrane potential falls.
- 4 Hyperpolarization (Undershoot):**
 - K⁺ channels close slowly → K⁺ continues to leave.
 - Membrane potential drops below resting (~-80 to -90 mV).
- 5 Return to Resting Potential:**
 - K⁺ channels close. Na⁺/K⁺ pump and leak channels restore the resting potential (~-70 mV).

3. REFRACTORY PERIODS

- ★ **Absolute Refractory Period:** From start of depolarization through early repolarization. Most voltage-gated Na⁺ channels are inactivated → Cannot fire another AP.
- ★ **Relative Refractory Period:** During late repolarization and hyperpolarization. Some Na⁺ channels recovered but membrane is hyperpolarized → A stronger stimulus is required to fire another AP.



Memory Trick: Na⁺ IN (Depolarization) before K⁺ OUT (Repolarization). "In Before Out!"



a) The Refractory Period

- ❑ While the electrical potential across the membrane is returning from its peak toward the resting point, it is still above the threshold.
- ❑ Why doesn't the cell produce another action potential during this period? (If it did, of course, it would go into a permanent repetition of one action potential after another.)
- ❑ Immediately after an action potential, the cell is in a **refractory period** during which it resists the production of further action potentials.
- ❑ In the first part of this period, the **absolute refractory period**, the membrane cannot produce an action potential, regardless of the stimulation.
- ❑ During the second part, the **relative refractory period**, a stronger than usual stimulus is necessary to initiate an action potential.
- ❑ Example: Toilet analogy - there is a short time right after you flush a toilet when you cannot make it flush again—an absolute refractory period. Then follows a period when it is possible but difficult to flush it again—a relative refractory period—before it returns to normal.
- ❑ The refractory period has two mechanisms: The sodium channels are closed, and potassium is flowing out of the cell at a faster than usual rate.

b) The All-or-None Law

- When the voltage across an axon membrane reaches the threshold, voltage-gated sodium channels open wide enough to let sodium ions enter, and the incoming sodium depolarizes the membrane enough to produce an action potential.
- For a given neuron, all action potentials are approximately equal in amplitude (intensity) and velocity (speed).
- The **all-or-none law** states that the amplitude and velocity of an action potential are independent of the intensity of the stimulus that initiated it, provided that the stimulus reaches the threshold.
- By analogy, imagine flushing a toilet: You have to make a press of at least a certain strength (the threshold), but pressing harder does not make the toilet flush faster or more vigorously.
- Similarly, once the stimulus reaches threshold, the action potential is full-sized regardless of stimulus strength. A stronger stimulus does not make it bigger or faster; it may only make the neuron fire more frequently.

Propagation of the Action Potential



How does the action potential move down the axon?

- Remember, it is important for axons to convey impulses without any loss of strength over distance.
- In a motor neuron, an action potential begins on the axon hillock.
- During the action potential, sodium ions enter a point on the axon.
- Temporarily, that spot is positively charged in comparison with neighboring areas along the axon.
- The positive ions flow within the axon to neighboring regions in both directions along the axon.
- The positive charges slightly depolarize the next area of the membrane, causing it to reach its threshold and open its voltage-gated sodium channels. Therefore, the membrane regenerates the action potential at that point.
- In this manner, the action potential travels along the axon
- The term **propagation of the action potential** describes the transmission of an action potential down an axon.
- Does action potential beginning at one point becomes weaker as they travel?
- The action potential gives birth to a new action potential at each point along the axon. As a result, the action potential is just as strong at the end of the axon as it was at the beginning, it doesn't become weaker when they travel from one area of axon to the other.
- Why the action potential travels in only one direction, even though the electrical charge spreads in both directions?
- Action potential cannot move backward because the areas that have just fired are still in their refractory period and cannot generate another action potential.

<https://youtu.be/TJq5E1K3ggg?si=x0yw2ckaokDcDbyL>

The Myelin Sheath and Saltatory Conduction



- ❑ **Myelin sheath** is an insulating material composed of fats and proteins.
- ❑ **Myelinated axons**, found only in vertebrates, are covered with layers of fats and proteins.
- ❑ The myelin sheath is interrupted periodically by short sections of axon called **nodes of Ranvier**, each one about 1 micrometer wide.
- ❑ Suppose an action potential starts at the axon hillock and propagates along the axon until it reaches the first myelin segment.
- ❑ The action potential cannot regenerate along the membrane between nodes because sodium channels are virtually absent between nodes.
- ❑ After an action potential occurs at a node, sodium ions enter the axon and diffuse within the axon, pushing a chain of positive ions along the axon to the next node, where they regenerate the action potential.
- ❑ This flow of ions is considerably faster than the regeneration of an action potential at each point along the axon.
- ❑ The jumping of action potentials from node to node is referred to as **saltatory conduction**, from the Latin word **saltare**, meaning “to jump.”
- ❑ Advantages of saltatory conduction
 - Provide rapid conduction of impulses
 - Conserve energy

The Myelin Sheath and Saltatory Conduction (Cont.)



What about unmyelinated axons?

- An axon that never had a myelin sheath **conducts impulses slowly but steadily**

What about axons that lost its myelin due to multiple sclerosis? (n autoimmune disease in which the immune system damages myelin, disrupting the normal transmission of action potentials in the nervous system)

- If the axon loses its myelin, it lacks sodium channels in the areas previously covered with myelin, and most action potentials die out between one node and the next.
- People with multiple sclerosis suffer a variety of impairments, ranging from visual impairments to poor muscle coordination.

Local Neurons



- Axons produce action potentials.
- However, many small neurons have no axon.
- Neurons without an axon exchange information with only their closest neighbors.
- We therefore call them **local neurons**.
- Because they do not have an axon, they do not follow the all-or-none law.
- When a local neuron receives information from other neurons, it has a **graded potential**, a membrane potential that varies in magnitude (size) in proportion to the intensity of the stimulus.
- That is, the **strength of the electrical change depends on the strength of the stimulus**.
- Eg: if strength of the stimulus is weak = change in electrical charge inside the neuron will also be weak = weak graded potential = potential travel only for a shorter distance.
- Stronger stimulus → bigger graded potential → spreads farther
- The change in membrane potential is conducted to adjacent areas of the cell, in all directions, **gradually decaying** as it travels.
- Those various areas of the cell contact other neurons, which they excite or inhibit through synapses
- Local neurons are difficult to study because it is almost impossible to insert an electrode into a tiny cell without damaging it.

THANK YOU!

REFERENCE

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